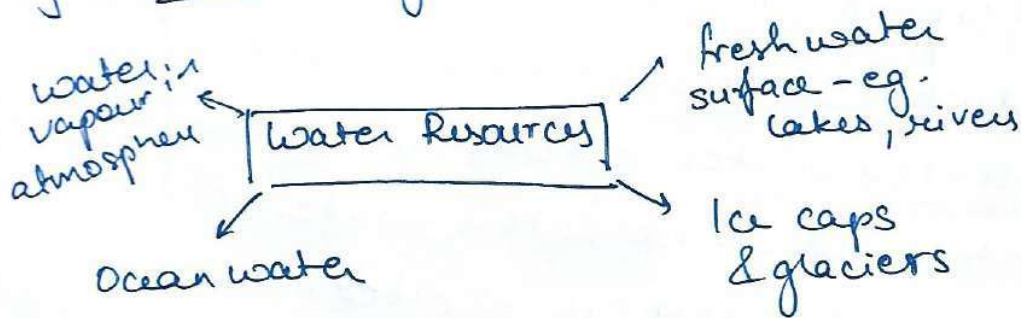




Q.15) Water resources are sensitive to variations in climatic patterns. In this context, elaborate the changes resulting from climate change and global warming on different water resources at the local and global scale. (15 Marks, 250 Words)

जल संसाधन जलवायु पैटर्न में बदलाव के प्रति संवेदनशील हैं। इस संदर्भ में, स्थानीय और वैश्विक स्तर पर विभिन्न जल संसाधनों पर जलवायु परिवर्तन और ग्लोबल वार्मिंग के परिणामस्वरूप होने वाले परिवर्तनों पर विस्तृत चर्चा करें। (15 अंक, 250 शब्द)

Water resource is the most vital life giving resource on Earth, which is facing existential challenge in the wake of climate change.



Water in all these resources is affected by climate change & global warming as:

① Ocean water

- Climate change causes the dangerous trika of ocean acidification, de-oxygenation and sea water rise.

This creates hypoxic regions in ocean water. eg. Baltic Sea dead zone

- Sea water rise creates problems of salt water infusion in ground water. eg. Vishakapatnam - sea erosion

② Ice Caps & Glaciers

- Climate change is causing melting of ice-caps & glaciers leading to:
 1. Reduction in freshwater resource.
 2. Polluted water eg. DDT etc. which is trapped in ice enters water.
 3. Source of rivers is effected.
eg. Melting of ice-caps in Antarctica.
Third pole in Himalaya being melted.

③ Surface fresh water

It is leading to drying out of freshwater resources due to rise in global temperatures

eg. Shrinking of the Lake Ural
pollution in Bangalore lakes
caused by global warming.

(4) water vapour, soil water, ground water

Climate change is affecting water
in these sources due to:

- erratic rainfall, lesser replenishment
- Aggravating natural disasters
like drought, heatwaves, etc
eg. Drought in Latur, Maharashtra.
Drought conditions in Cape Town, South Africa

Way forward

- Curb global warming by following
Paris Agreement commitments.
- Curbing water pollution at source.
- Encouraging re-use of water.

Feedback (For OFFICE use only)

Structure		Content	
Question Interpretation		Total :	